Noor Khial

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Summary

AI/ML engineer and researcher with experience building and deploying machine learning systems for **customs and logistics** operations in Qatar. At Malomatia I develop AI services for the Authority of Customs, integrating models into platforms already in daily production use and automating processes that were previously manual.

My research focuses on **continual learning** and **meta-learning**: how can a learning system sustain its performance after deployment, as conditions shift in ways not known in advance? In practice, I train models on historical production data and adapt them to live traffic, deploying into production through **generative AI**, continual learning, meta-learning, and other approaches. My most recent interest is **multi-agent systems**, given their potential across customs problems and beyond.

Experience

09/2025–Present **Part-Time Researcher**, *Qatar University*, Qatar
- Applying online, continual and meta-learning theory to decision systems that must stay reliable after deployment.
- Online-learning framework combining stochastic and adversarial rates for target detection, under QRDI-funded research.

05/2025–Present **AI/ML Engineer**, *Malomatia*, Qatar
- Design and deploy AI services across two of the three customs AI pillars, tariff classification and goods valuation, integrating models into systems already in daily production use.
- Automate processes that were previously manual: HS code classification, goods valuation, declaration drafting and duty estimation.
- Own the full chain end to end, from Oracle extraction and OCR through preprocessing and feature engineering to transformer fine-tuning, meta-learning models and serving on Azure.
- Classification is live in production at **92% accuracy**, an **80% efficiency improvement** over manual work; valuation is in staging at **71% accuracy**, **60%** above the manual baseline. This work took 1st place in the Malomatia AI Contest 2025 and 3rd in the AI Hackathon 2026.
- **Tool chain**: Oracle, Azure (Document Intelligence, AI Search, OpenAI, Functions), PyTorch, Hugging Face, LangChain, FastAPI, Docker, SQL.

02/2025–04/2025 **AI Engineer**, *Applab*, Qatar
- Built multi-agent AI chatbots for Qatari national institutes on Microsoft Azure, chaining Azure OpenAI Service with LangChain orchestration and retrieval over a vector index.

01/2023–01/2025 **Graduate Research Assistant**, *Qatar University*, Qatar
- Designed online-learning and deep reinforcement-learning algorithms for UAV navigation in non-stationary and adversarial environments, with regret-based guarantees.
- Contributed to multiple funded research projects applying meta-learning to autonomous systems.
- Published two journal papers in top-ranked AI journals and two peer-reviewed conference papers.

09/2020–05/2021 **Academic Tutor**, *Qatar University*, Qatar
- Tutored undergraduate students in Calculus II and III and in Probability and Statistics.

Education

01/2023–12/2024 **Master of Science in Computing**, *Qatar University*, Qatar, GPA: 3.77/4.00
- Thesis: "Smart Path Planning for Mobile Multi-Target Visitation in UAV Missions in Stochastic and Adversarial Environments"
- Supervisor: Prof. Amr Mohamed, Head of the CSE Department, Qatar University.
- "Research Distinguished" Award. Funded by a Graduate Research Assistantship (tuition and stipend).
- Coursework: Algorithms, Deep Learning, Information Retrieval, Computer Vision.

01/2018–05/2022 **Bachelor of Science in Computer Engineering**, *Qatar University*, Qatar, GPA: 3.50/4.00
- Senior Project: onboard computer and transceiver subsystem for QUbeSat, the first CubeSat launched by Qatar.

- Supervisor: Prof. Tamer Khattab, Head of the Intelligent Information Processing Lab (IIPL), Qatar University.
- "Amiri Diwan" Scholarship (merit-based, full tuition).
- Coursework: Probability and Statistics, Discrete Mathematics, Machine Learning.

Publications

Journal Papers

2026 **Khial, N.**, Allahham, M.S., Mhaisen, N., Ismail, L., Mabrok, M., and Mohamed, A., "DRONE-RL: Dynamic reinforcement learning for online navigation of UAVs in evolving environments," *Knowledge-Based Systems*, Elsevier, vol. 334, p. 115147.

2025 **Khial, N.**, Mhaisen, N., Mabrok, M., and Mohamed, A., "An online learning framework for UAV search mission in adversarial environments," *Expert Systems with Applications*, Elsevier, vol. 267, p. 126136.

Conference Papers

2025 **Khial, N.**, Mhaisen, N., Ismail, L., Mabrok, M., and Mohamed, A., "Multi-Target Path Planning with Probabilistic Detection in Cluttered Environments," *IEEE International Conference on Communications (ICC)*, pp. 1298–1303.

Under Submission

Pending **Khial, N.**, Mhaisen, N., and Mohamed, A., "Multi-Regime Policy Selection for Online Search Using Stochastically Extended Adversarial Model," submitted to the *IEEE Internet of Things Journal*.

Open-Source Research Software

Arabic Information Retrieval System
- Semantic search for Arabic document retrieval, combining FAISS with the mContriever model. *Python*, *FAISS*, *PyTorch*, *Hugging Face*.

UAV Navigation in Adversarial Environments
- Deep reinforcement learning with a custom simulation environment for multi-target tracking. *Python*, *TensorFlow*, *OpenAI Gym*.

Skills

Research	Continual learning, meta-learning, online learning, deep reinforcement learning, multi-agent systems, generative AI.
ML and DL	PyTorch, TensorFlow, Hugging Face Transformers, BERT and AraBERT, SBERT, FAISS, scikit-learn.
LLM and agents	LangChain, Azure OpenAI, retrieval-augmented generation, vector search, multi-agent orchestration.
Cloud	Azure Functions, Azure Document Intelligence, Azure AI Search, Blob Storage, FastAPI, Flask, Docker.
Data	Oracle, SQL, pandas, OCR extraction pipelines, feature engineering.
Programming	Python, C, JavaScript, MATLAB.
Languages	Arabic (native), English (fluent).

Awards

2026 "3rd Place" Malomatia AI Hackathon, for IntelliLogistiX.

2025 "1st Place" Malomatia AI Contest, for HSsify Engine.

2025 "Research Distinguished" Award, Qatar University.

2023–2024 Graduate Research Assistantship, Qatar University.

2018–2022 "Amiri Diwan" Scholarship, merit-based full tuition.

2019–2022 Dean's List, College of Engineering, Qatar University.

2022 QU Undergraduate Startups Pre-Incubation Program, Qatar University.